A NOVEL RELIABILITY AWARE PATCH SELECTION BASED FEW SHOT LEARNING FOR TARGET RECOGNITION AND OPEN-SET IDENTIFICATION IN ILL POSED SAR IMAGES

A thesis submitted in partial fulfilment of the requirements
for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

by

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April 2026

Evaluation Sheet

Title of the Project:

A Novel Reliability Aware Patch Selection Based Few Shot Learning for Target Recognition and Open-Set Identification in Ill Posed SAR Images

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Declaration

We, the undersigned students, hereby declare that the material presented in the Project Report titled *"A Novel Reliability Aware Patch Selection Based Few Shot Learning for Target Recognition and Open-Set Identification in Ill Posed SAR Images"* represents original work carried out by us in the Department of Computer Science and Engineering at the Indian Institute of Information Technology Design and Manufacturing, Kurnool during the year 2025–2026. With our signatures, we certify that:

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- We have explicitly acknowledged all collaborative research and discussions.
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In my capacity as supervisor of the above-mentioned work, I certify that the work presented in this Report is carried out under my supervision and is worthy of consideration for the requirements of B.Tech Project work.

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Abstract

Synthetic Aperture Radar (SAR) is a critical technology for all-weather, day-and-night Earth observation, with applications in environmental monitoring, defence, maritime safety, and urban planning. However, SAR target recognition remains fundamentally ill-posed due to limited labeled data, severe class imbalance, and open-set conditions where unseen targets frequently occur. Conventional deep learning methods, which assume abundant balanced data and closed label spaces, often fail under these constraints.

This thesis proposes a unified framework based on a novel patch-wise manifold learning architecture.Our main idea is to mix a residual multi-scale encoder with a filtering method that uses entropy to sort out image patches. This ensures we can focus on relevant features in the images while minimizing the impact of noise. After that, the filtered features will be used to construct prototypes for each class using a dependency matrix that captures rela- tionships between different feature channels. This approach is more advanced compared to the straightforward averaging approach. Training the model requires a loss function that is a combination of contrastive learning, prototype clustering, and a classification term. The contrastive learning loss focuses on learning by comparison, while the prototype cluster- ing loss considers the prototypes of each class and minimizes the distance between sam- ples within each class while maximizing the distances between samples from different classes. The classification term incorporates sample weights based on their frequencies. This loss function will ensure that samples in the same class are pulled closer while sam- ples in different classes are repelled from each other. For the unseen object detection task, we simply apply the nearest neighbor classification based on distances. This approach will also have adaptive thresholds for each episode.

Acknowledgements

We are grateful to **Dr. Shounak Chakraborty**, Assistant Professor, Department of Computer Science and Engineering, for his valuable suggestions and guidance during the process of this project. He provided valuable technical information and suggestions that helped us solve some of the problems faced while researching and made us concentrate on the goal.

We are thankful to **Vishnu Meher Vemulapalli** and **Anisha Chakravorty**, students who work under Dr. Shounak Chakraborty, for their co-operation and sharing their knowledge with us. They made our learning process easier and efficient, which helped us complete this project successfully. We also thank Indian Institute of Information Technology Design and Manufacturing, Kurnool, for providing an ideal environment for us to learn and do research. We could not have completed this project without their help.

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Abbreviations

CNN	Convolutional Neural Network
SAR	Synthetic Aperture Radar
FSL	Few-Shot Learning
OSR	Open-Set Recognition
Grad-CAM	Gradient-weighted Class Activation Mapping
mIoU	mean Intersection over Union
MSTAR	Moving and Stationary Target Acquisition and Recognition
SAMPLE	Synthetic and Measured Paired Labeled Experiment
DLRSD	Deep Learning Remote Sensing Dataset
MA-UNet	Multi-Attention U-Net
XAI	Explainable Artificial Intelligence

Chapter 1

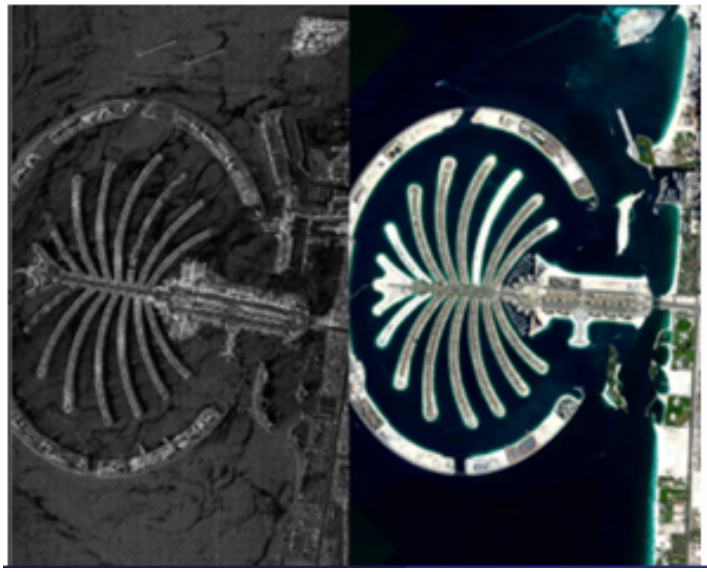
Introduction

1.1 Background and Context

The advent of Synthetic Aperture Radar (SAR) has fundamentally transformed the landscape of Earth observation, introducing capabilities that extend far beyond the limitations of conventional optical imaging systems. Unlike passive optical sensors that rely on solar illumination and favourable atmospheric conditions, SAR systems actively transmit electromagnetic pulses toward the Earth's surface and record the backscattered energy, thereby generating high-resolution imagery regardless of weather conditions or time of day [1, 2]. This all-weather, day-and-night imaging capability renders SAR indispensable for a wide spectrum of applications, including environmental monitoring, disaster assessment and response, defence and security surveillance, maritime safety and vessel tracking, urban planning and infrastructure management, and agricultural monitoring and crop assessment [1, 3].

SAR images look different from regular photos because of how radar works. These images form when radar signals bounce off the ground. The final picture depends on the angle of the signal, its polarization, and how rough the surface is.

Because radar sees things differently, it can show details about materials and structures that regular cameras miss. However, this also makes the images harder for computers to read. Radar images often contain "speckle," which is a grainy blur that makes it difficult to tell objects apart. Moreover, the appearance of the same target is bound to vary if there is a change in weather conditions or due to any other reason. There could be cases where two entirely different targets might appear identical to each other.



Palm Island Dubai, UAE.
(a) Optical Satellite image
(Maxar Technologies)
(b) SAR Satellite image
(Capella Space)

Figure 1.1: Palm Island Dubai, UAE (a) Optical Image (b) SAR Image

1.2 The Ill-Posed Nature of SAR Target Recognition

In our attempt to employ deep learning techniques for detecting targets from SAR images, we face a number of interconnected bottlenecks. Each of these bottlenecks is a significant obstacle by itself, yet they all seem to be interwoven together, making the problem even more complicated and difficult to resolve.

Scarcity of Labeled Data: Undoubtedly, one of the major obstacles in SAR imagery analy- sis is the absence of annotated data sets. Specifically, to achieve accurate labeling, some special knowledge is required, which is not commonly available. Moreover, a big part of SAR image datasets remains private for security reasons. Acquisition is costly and challenging, especially the organization of extensive campaigns to obtain ground truth datasets. Public SAR benchmarks are rather small from the standpoint of modern deep learning. Thus, MSTAR benchmark, which is commonly used in the research community, contains several thousands of images, whereas ImageNet dataset comprises millions of optical images. [5].

Class Imbalance: Even with a limited amount of information, the problem becomes particularly difficult when certain groups of data are significantly larger than other groups [4, 6]. In practical applications, such as in the case of radar, it becomes evident that the sought-for object is not evenly distributed. As a result, a data set can contain one thousand occurrences of the standard T-72 tank, while only twenty-five occurrences of an uncommon electronic warfare vehicle. Such a data distribution affects the way the information appears to be structured. Consequently, the machine learning algorithm, based on such a data set, will perform well at recognizing objects frequently observed in training. Rare cases tend to be ignored by such systems, and even solutions

that are commonly used, such as multiplying rare occurrences and generating new samples, do not help solve the issue [4, 5]. Instead, these techniques merely amplify the initial mistakes while failing to account for the variety of rare cases, despite the latter being more diverse than the number of occurrences suggests.

Open-Set Conditions: However, an extremely large problem, one that is rarely considered in research works, is that as soon as SAR is applied to any practical task, we are faced with something completely new [7, 8].During experiments and evaluation, we are dealing with a set of object types. Therefore, each image must belong to one of them, and the model should be able to recognize which one. However, when working with real data, we often encounter completely novel objects such as a new type of vehicle, weather phenomena that were not previously encountered, or simply anything unusual that cannot be classified. In this situation, if the algorithm knows a specific set of object types in advance, then it will consistently misclassify the sample under consideration, assigning it to one of the known classes. Such misclassification can lead to catastrophic consequences in critical applications, including threat detection, anomalous event detection in maritime traffic, or disaster response. It means threats can slip by, or people in charge might miss really important stuff [1, 3].

1.3 Research Questions

This report asks the following questions:

Primary Research Question: How is it possible to create one system that works well on three important problems at once, like having too little data, unequal distribution among different groups, and working with unknown targets in the SAR target recognition problem?

Subsidiary Research Questions: The questions posed are designed to help define the problem, analyze the current solution(s), and determine the gap that the proposed solution addresses.

1. What would be the formal description of the challenge posed by noise and unreliability in patch representations in SAR images? How can we evaluate patch stability to enable the selection of patches that form reliable prototypes especially for minority classes?

2. What are the shortcomings of the existing prototype-based techniques that consider centroids alone, and what should be done to enhance class prototypes by considering the relationships between the feature channels?

3. What techniques could be utilized for improving these three methods together, such that their weaknesses are combined into strengths resulting in the creation

of a feature space with higher intra-class compactness and inter-class separation using little and imbalanced data?

4. What is the solution to the issue of open-set classification in SAR images using the distance-based rejector method without additional unknown support samples or artificially generated unknown samples?

1.4 Contributions of the Thesis

This thesis makes the following novel contributions to the field of SAR target recognition:

1. **Reliability-Aware Patch Learning Framework:** A novel patch-wise learning scheme for SAR imagery that integrates entropy-guided patch selection to ensure that only discriminative, imbalance-resilient, and structurally stable regions contribute to feature embedding.

2. **Dependency Matrix-Based Prototype Construction:** A novel prototype construction strategy that operates on a dependency matrix manifold, encoding channel-wise relational structures into class prototypes.

3. **Unified Episodic Learning Framework:** A novel unified learning paradigm that integrates instance-level contrastive alignment, prototype-driven clustering, and imbalance-aware classification within a single episodic optimization framework.

4. **Adaptive Open-Set Rejection Mechanism:** A distance-based open-set recognition mechanism that leverages the learned embedding space without relying on auxiliary support samples or synthetic unknown data.

1.5 Thesis Organisation

The remainder of this thesis is organized as follows:

- **Chapter 1** introduces the problem domain of SAR target recognition and outlines the key challenges, including data scarcity, class imbalance, and open-set conditions. It presents the problem definition, discusses existing approaches and their limitations, and formulates the research questions and contributions of the thesis.

- **Chapter 2** gives a full overview of related work in SAR target recognition, few-shot learning, fixing class imbalance, open-set recognition, and feature extraction based on attention.

- **Chapter 3** gives a detailed explanation of the proposed framework. It covers problem formulation, architectural design, entropy-guided patch selection, dependency matrix prototype construction, multi-component loss optimisation, and open-set rejection mechanisms.
- **Chapter 4** talks about the datasets, evaluation protocols, implementation details, hyperparameter settings, and baseline methods that were used in the experimental validation.
- **Chapter 5** presents thorough experimental findings, encompassing ablation studies, quantitative assessments against cutting-edge techniques, and qualitative evaluations of learned representations via t-SNE visualisations.
- **Chapter 6** critically assesses the outcomes, contextualising the findings within the framework of the research questions and literature review, while addressing the limitations of the present study.
- **Chapter 7** summarises the thesis's contributions and suggests areas for future research.

Chapter 2

Literature Review

2.1 Introduction

This chapter offers an extensive analysis of the literature related to SAR target recognition, few-shot learning, class imbalance mitigation, open-set recognition, and attention-based feature extraction. The review contextualises the proposed framework within the extensive research domain, delineates deficiencies in current methodologies, and formulates the theoretical underpinnings that support this study.

2.2 Synthetic Aperture Radar: Principles and Challenges

2.2.1 Physical Principles of SAR Imaging

Imaging by SAR depends on the acquisition of coherent signals, which then undergo processing that leads to the creation of a virtual aperture of very high resolution that is achieved irrespective of the distance of imaging [1]. Amplitude and phase values acquired are then processed to form spatial images.

The backscattered signal captured in SAR imagery is contingent upon various interacting factors: the dielectric properties of the target material, which dictate reflectivity; surface roughness in relation to the radar wavelength, which impacts scattering mechanisms; target geometry and orientation, which influence the strength and direction of returns; and environmental conditions, including soil moisture and vegetation cover. [2]. These dependencies give SAR images a lot of information, but they also make them very different, which makes it hard for computers to understand them.

2.2.2 Distinctive Characteristics of SAR Data

SAR imagery has a number of unique features that set it apart from optical imagery and make it hard for deep learning applications to use. [2]:

- **Speckle Noise:** Speckle is a granular interference pattern that happens when returns from multiple scatterers within a resolution cell are coherently summed. SAR images are always affected by this.
- **Geometric Distortions:**The side-looking geometry of SAR acquisition introduces characteristic geometric distortions, including foreshortening, layover, and shadowing.
- **Polarimetric Diversity:** Present-day SAR sensors usually use more than one polarization, which gives extra information on scattering mechanisms.
- **Interferometric Capabilities:** InSAR is based on using observations taken from several locations in order to produce digital elevation models (DEMs).

2.3 Few-Shot Learning in SAR Applications

2.3.1 Foundations of Few-Shot Learning

The few-shot learning paradigm addresses the situation of having to learn in the scenario of very few labeled instances. The paradigm can be expressed as N-way K-shot classification, where the model must classify things into N classes despite being presented with only K labeled instances. [16]. This field has changed a lot over time, going from methods based on measurements (like Prototypical Networks) to ones focused on making things work better through optimization (like MAML), and now we have methods that use a kind of ”hallucination” approach.

2.3.2 Few-Shot Learning for SAR Target Recognition

Using few-shot learning for SAR target recognition is getting a lot more attention these days, mainly because we still don’t have much labeled SAR data [14, 16]. At first, researchers mostly took general few-shot methods and simply tried them out on SAR images, often without truly thinking about the unique things that make SAR different. Then, meta-learning approaches, for example, Meta-FSCIL [11] , came along, which can adaptively learn the best ways to separate different classes. Other methods, like position-aware graph neural networks [13] **use graphs to model the structural connections** between known and unknown samples .Contrastive and prototype frameworks such as MCL-DMM [14] and MHD-ProtoNet [15] have demonstrated promising results.

2.4 Class Imbalance in SAR Target Recognition

2.4.1 Nature and Consequences of Class Imbalance

Class imbalance in SAR target recognition stems from the inherent non-uniformity of real-world target distributions [4, 6]. This imbalance leads to representational bias, prototype deformation, and gradient dominance.

2.4.2 Conventional Mitigation Strategies

Resampling (oversampling minority classes or undersampling majority classes) and synthetic data generation are two examples of data-level approaches [5]. Cost-sensitive learning, class-weighted loss functions, and focal loss are all examples of algorithm-level approaches. There are a lot of different strategies that work together in hybrid approaches.

2.4.3 Imbalance-Aware Methods for SAR

Recent research has commenced to tackle class imbalance specifically within SAR contexts [10]. Constraint and optimization-based methods either change the weights of gradients or add class-wise constraints. Data-centric strategies investigate generative augmentation through GANs [5] and entropy-driven resampling.

2.5 Open-Set Recognition

2.5.1 Theoretical Foundations

Open-set recognition deals with the real-world situation where test samples might belong to classes that weren't seen during training [7, 8].OSR is different from closed-set classification in that it needs models to both correctly classify known classes and reliably reject unknown ones.

2.5.2 Open-Set Recognition Approaches

Several paradigms have developed: discriminative approaches (training to differentiate known from unknown), generative approaches (modelling the probability of known classes), prototype-based approaches (distance-based rejection), and evidential approaches (quantification of uncertainty).

2.5.3 Open-Set Recognition in SAR

How SAR analysis works in real life, often encountering unknown things, has really sparked a lot of research interest [7–9].With these ”threshold-free open-set learning networks,” [8] we no longer need to manually set those limits for rejecting data . Capule network methods [7] ecan pull out details about an object’s position and then compare those details using certain techniques. And Incremental learning frameworks [9] ialso include ways to find important features by looking at how similar their structures are.

2.6 Attention Mechanisms and Multi-Scale Feature Extraction

2.6.1 Attention in Deep Learning

Attention mechanisms help computer models zero in on the important parts of the data they receive, letting them ignore everything else that isn’t useful. You’ll find different types of attention [13].For example, spatial attention highlights the crucial areas in an image. Channel attention focuses on giving different importance levels to various features. And then there’s self-attention, which is all about figuring out how different pieces of information relate to each other.

2.6.2 Multi-Scale Feature Extraction

When we look at SAR images, it’s really important to extract features at different scales. [8,13].This is because the objects we’re trying to find can be various sizes, and the images themselves might have different levels of detail. This is because the objects we’re trying to find can be various sizes, and the images themselves might have different levels of detail. Some common ways to do this include using feature pyramid networks (FPNs), atrous spatial pyramid pooling (ASPP), or an encoder-decoder design that includes skip connections.

2.7 Identified Research Gaps

There are many defeciencies in the current research:

- Lack of unified frameworks that deal with data scarcity, imbalance, and openness at the same time.
- Insufficient focus on patch-level reliability during prototype development.
- Limited use of internal feature structure beyond centroid prototypes.

- Dependence on synthetic unknowns in numerous open-set methodologies.

2.8 Summary

The above chapter gives an overview of the literature review on SAR target recognition through classification of existing studies in major categories like few-shot learning methodologies, approaches to solve class imbalance problems, open set recognition techniques, and attention-based approaches for feature extraction. Some prominent examples from each of these categories have been discussed to identify their pros and cons. The gap analysis shows that there is a need for an approach that can tackle all the three challenges mentioned above at once, but existing approaches solve each challenge individually.

Chapter 3

Methodology

3.1 Introduction

This chapter has looked at the literature on SAR target recognition, few-shot learning, fixing class imbalance, open-set recognition, and feature extraction based on attention. The identified deficiencies necessitate the unified framework proposed in this thesis.

3.2 Problem Formulation

3.2.1 Notation and Definitions

Let the training dataset be denoted as $D_{base} = \{(x_i, y_i)\}_{i=1}^{M}$, where x_i represents a SAR image and y_i denotes its corresponding class label from a set of known base categories $C_{base} = \{c_1, c_2, \ldots, c_n\}$. The imbalance ratio is defined as:

$$i_d = \frac{|D(c_{min})|}{|D(c_{max})|}$$

where $|D(c_{min})|$ represents the number of samples of the rarest class and $|D(c_{max})|$ the most frequent class.

3.2.2 Testing Scenarios

Closed-Set Few-Shot Classification: A novel dataset $D_{new} = S_{new} \cup Q_{new}$ is presented with novel classes C_{new} disjoint from C_{base}.

Open-Set Recognition: The label space extends to $C_{new} \cup \{-1\}$, where -1 denotes unknown classes.

3.3 Residual Contextual Feature Encoding

3.3.1 Architectural Design Choices

The feature encoding phase employs a residual convolutional architecture designed specifically for SAR imagery, with residual connections mitigating vanishing gradients and facilitating gradient flow.

3.3.2 Data Augmentation for SAR Stability

Geometric augmentations include rotation ($\pm 10°$), scaling ($\pm 5\%$), translation (± 5 pixels), and synthetic speckle addition.

3.3.3 Feature Map Extraction

Both the original image x and its augmented variant x' are processed through a shared residual convolutional network parameterized by θ, yielding deep spatial feature maps:

$$f_\theta(x), f_\theta(x') \in \mathbb{R}^{H \times W \times C}$$

3.3.4 Patch-Wise Decomposition

The feature map is partitioned into $G \times G$ non-overlapping spatial regions:

$$f_\theta(x) = \{p_i\}_{i=1}^{G^2}$$

Each patch is transformed into a patch-wise vector:

$$\mathbf{z}_i = \phi(p_i) \in \mathbb{R}^d$$

3.3.5 Entropy-Guided Patch Selection

Each patch vector is normalized into an activation distribution:

$$\mathbf{a}_i(k) = \frac{\exp(\mathbf{z}_i(k))}{\sum_{u=1}^{d} \exp(\mathbf{z}_i(u))}, \quad k = 1, \ldots, d$$

where $\mathbf{z}_i \in \mathbb{R}^d$ denotes the feature vector corresponding to the i-th patch, and $\mathbf{a}_i$ represents the normalized activation (probability) distribution obtained by applying the softmax function over $\mathbf{z}_i$. Here, k denotes the index of a specific feature dimension, u is the summation index over all feature dimensions, and d is the dimensionality of the feature

vector. Uncertainty is quantified using Shannon entropy:

$$E_i = -\sum_{k=1}^{d} \mathbf{a}_i(k) \log \mathbf{a}_i(k)$$

A binary mask suppresses unreliable patches based on entropy threshold τ:

$$m_i = \begin{cases} 1, & E_i < \tau \\ 0, & E_i \geq \tau \end{cases}$$

yielding filtered descriptors $\tilde{\mathbf{z}}_i = m_i \mathbf{z}_i$.

3.3.6 Global Context Preservation

A global descriptor is obtained by spatial average pooling:

$$\mathbf{z}_g = \text{GlobalAvgPool}(f_\theta(x)) \in \mathbb{R}^d$$

3.3.7 Multi-Scale Feature Fusion

The multi-scale representation combines global and local features:

$$\mathbf{z}(x) = \psi\left(\mathbf{z}_g, \{\tilde{\mathbf{z}}_i\}_{i=1}^{G^2}\right)$$

where $\psi(\cdot)$ denotes a feature fusion function that combines the global descriptor $\mathbf{z}_g \in \mathbb{R}^d$ with the set of filtered local descriptors $\{\tilde{\mathbf{z}}_i \in \mathbb{R}^d\}_{i=1}^{G^2}$ through concatenation followed by aggregation, and $G \times G$ represents the spatial grid of patches.

3.3.8 Instance-Level Contrastive Learning

The augmentation loss enforces consistency between each sample and its augmented counterpart:

$$\mathcal{L}_{aug} = \text{CE}\left(\cos\left(\mathbf{z}(x), \mathbf{z}(x')\right), y\right)$$

where $\cos(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$ denotes cosine similarity and y is the ground-truth label.

3.4 Prototype Embedding with Dependency Matrix

3.4.1 Transformation to Embedding Space

Support samples are transformed into an embedding space:

$$\mathbf{v}(x) = \eta(\mathbf{z}(x)) \in \mathbb{R}^d$$

3.4.2 Initial Prototype Formation

For each class, a preliminary prototype is computed as the mean of transformed embeddings:

$$\mathbf{v}_{c_i} = \frac{1}{|S_{c_i}|} \sum_{x \in S_{c_i}} \mathbf{v}(x)$$

3.4.3 Dependency Matrix Construction

Pairwise channel similarities are computed:

$$m_{ij} = \kappa(v_{c_i,i}, v_{c_i,j})$$

yielding a symmetric dependency matrix:

$$\mathbf{M}_{c_i} = [m_{ij}] \in \mathbb{R}^{d \times d}, \quad i, j = 1, \ldots, d$$

The upper triangular portion is extracted to form the structural prototype:

$$p_{c_i} = \text{Upper}(\mathbf{M}_{c_i})$$

3.4.4 Inference with Dependency Prototypes

A query sample is processed similarly to obtain $\mathbf{p}_q$, and classification is performed by nearest prototype:

$$c^* = \arg \min_{c_i \in C_{known}} d(\mathbf{p}_q, \mathbf{p}_{c_i})$$

3.5 Episodic Structural Learning

3.5.1 Prototype-Driven Cluster Discrimination

The intra-class clustering loss pulls queries toward their correct prototype:

$$\mathcal{L}_{\text{intra}} = -\log \frac{\exp\left(-d(\mathbf{p}_q, \mathbf{p}_{y_q})\right)}{\sum_{c_i \in \mathcal{C}_{\text{epi}}} \exp\left(-d(\mathbf{p}_q, \mathbf{p}_{c_i})\right)}$$

3.5.2 Instance-Level Constraint

Instance-level contrastive loss compares queries with all support samples:

$$\mathcal{L}_{\text{ins}} = \sum_{x_s \in S_{c_i}} \ell\left(d(\mathbf{p}_q, \mathbf{p}_{c_i}), \mathbb{I}(y_q = y_s)\right)$$

3.5.3 Combined Prototype Loss

The combined prototype loss is:

$$\mathcal{L}_{\text{pro}} = \eta \mathcal{L}_{\text{intra}} + \lambda \mathcal{L}_{\text{ins}}$$

3.5.4 Imbalance-Aware Classification Loss

Classification logits based on distances:

$$\text{Logits}_{c_i} = -d(\mathbf{p}_q, \mathbf{p}_{c_i})$$

Weighted cross-entropy with class-frequency dependent weights:

$$\mathcal{L}_{\text{cls}} = \text{CE}(\text{Logits}, y_q, w_{c_i})$$

where $w_{c_i} = 1/\log(1 + |D(c_i)|)$.

3.5.5 Multi-Component Optimization

The complete meta-training objective integrates all losses:

$$\mathcal{L}_{\text{episode}} = \alpha \mathcal{L}_{\text{aug}} + \beta \mathcal{L}_{\text{pro}} + \gamma \mathcal{L}_{\text{cls}}$$

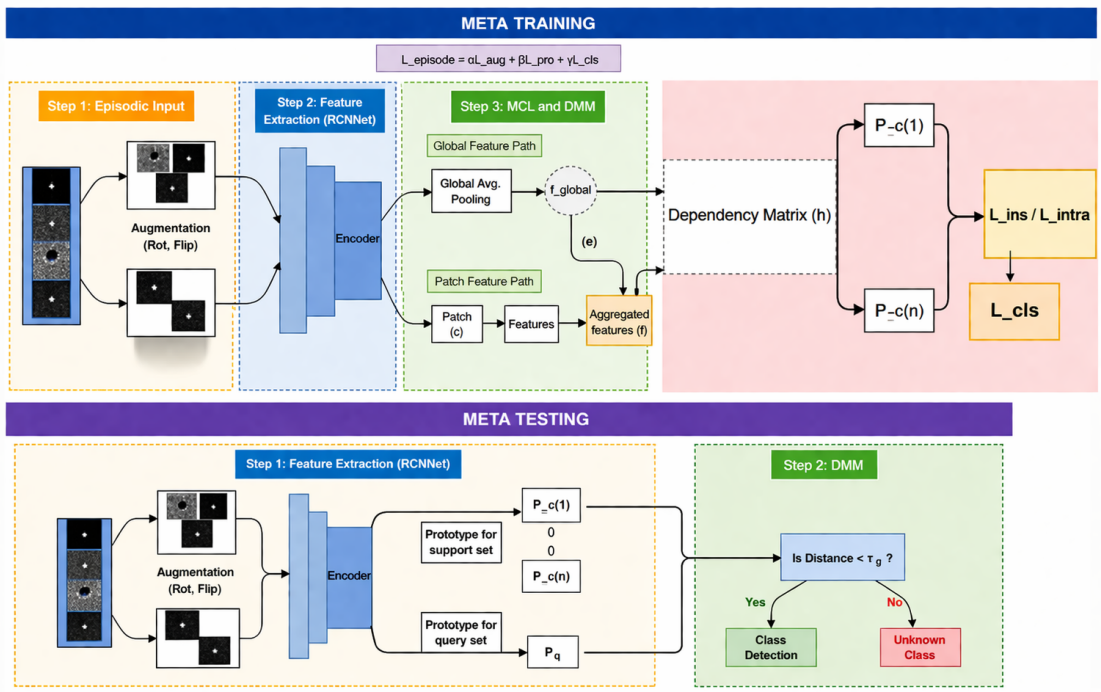


Figure 3.1: Our Proposed Workflow: (1) (Global+Patch) Feature Extraction → (2) MCL Training → (3) DMM Classification.

3.6 Open-Set Recognition Mechanism

3.6.1 Distance-Based Rejection Philosophy

Samples that do not belong to any known class lie farther from all class prototypes.

3.6.2 Adaptive Threshold Computation

For a query q, the minimum distance to known prototypes is:

$$d_{\min}(q) = \min_{s_i \in C_{new}} d(\mathbf{p}_q, \mathbf{p}_{s_i})$$

The prediction rule is:

$$\text{Prediction} = \begin{cases} \arg\min_{s_i} d(\mathbf{p}_q, \mathbf{p}_{s_i}), & d_{\min}(q) \leq \tau_{\text{global}} \\ -1, & d_{\min}(q) > \tau_{\text{global}} \end{cases}$$

with $\tau_{\text{global}} = \mu_d + k\sigma_d$, typically $k = 3$ where $\mathbf{p}_q \in \mathbb{R}^d$ denotes the embedding of the query sample q, $\mathbf{p}_{s_i} \in \mathbb{R}^d$ represents the prototype of the i-th known class, $d(\cdot, \cdot)$ is a distance metric (e.g., Euclidean distance), $d_{\min}(q) = \min_{s_i} d(\mathbf{p}_q, \mathbf{p}_{s_i})$ is the minimum distance between the query and all class prototypes, τ_{global} is the rejection threshold, μ_d and σ_d denote the mean and standard deviation of distances computed over known

classes, k is a scaling factor (typically $k = 3$), and a prediction of -1 indicates that the sample is classified as unknown.

3.7 Summary

This chapter has presented a comprehensive exposition of the proposed framework, including entropy-guided patch selection, dependency matrix prototype construction, multi-component episodic optimization, and distance-based open-set rejection.

Chapter 4

Implementation and Experimental Setup

4.1 Dataset Description

4.1.1 MSTAR Dataset

The Moving and Stationary Target Acquisition and Recognition (MSTAR) dataset contains X-band SAR imagery of ten military vehicle categories [17]. Seven categories are used as base classes (BMP2, BTR70, T72, BTR60, D7, T62, ZIL131), and three as novel classes (2S1, BRDM2, ZSU234). For open-set evaluation, ZSU234 is designated as unseen.

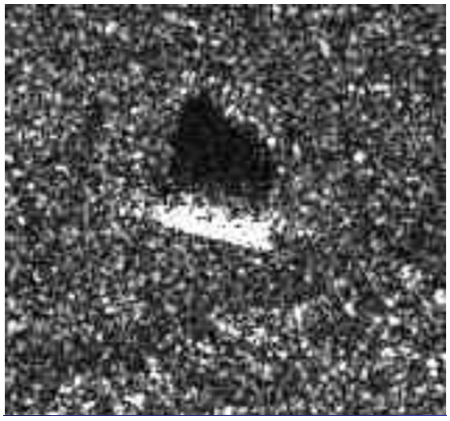


Figure 4.1: Sample SAR image from the MSTAR dataset showing a military vehicle target.

4.1.2 SAMPLE Dataset

The Synthetic and Measured Paired Labeled Experiment (SAMPLE) dataset contains both synthetic and measured imagery of ten vehicle types [18]. Seven classes form the base set, three are novel, and M1 is the unseen class for open-set evaluation.

4.1.3 Dataset Preprocessing

All images are converted to magnitude format, normalized to [0,1], resized to 128×128 pixels, and augmented online.

4.1.4 SAMPLE Dataset

Table 4.1: Details of the SAMPLE Dataset

Attribute	**Description**
Dataset Name	SAMPLE (Synthetic and Measured Paired Labeled Experiment)
Data Type	Synthetic and measured SAR imagery
Number of Classes	10 vehicle classes
Base Classes	7 classes
Novel Classes	3 classes
Open-set Class	M1 (unseen during training)
Image Resolution	128×128 pixels
Modality	Synthetic Aperture Radar (SAR)
Acquisition Source	Public benchmark dataset
Application Domain	Target recognition in SAR imagery

4.2 Implementation Details

4.2.1 Architecture Configuration

The residual encoder comprises four residual blocks with channel sizes 64, 128, 256, 512, totaling approximately 3.2M parameters. Patchification uses $G = 4$ (16 patches). Entropy threshold $\tau = 0.65$. Embedding dimension $d = 256$. Dependency matrix dimension $d(d-1)/2 = 32,640$. Similarity operator κ uses a learnable function.

4.2.2 Training Configuration

Meta-learning uses 3-way K-shot tasks with $K \in \{1, 3, 5, 7\}$. Episodes contain 3 classes, 5 support images per class, 15 query images total. Optimizer: Adam (lr=1e-4, weight decay=5e-4). Trained for 1000 episodes. Loss weights: $\alpha = 1.0, \beta = 0.5$ (with $\eta = 0.7, \lambda = 0.3$), $\gamma = 1.0$.

4.2.3 Open-Set Configuration

Rejection threshold $\tau_{\text{global}} = \mu_d + 3\sigma_d$, computed per test episode.

4.3 Baseline Methods

Five few-shot learning methods are compared: ConvT [16], Meta-FSCIL [11], Position-Aware GNN [13], MHD-ProtoNet [15], MCL-DMM [14]. For open-set, three additional methods: Incremental OSR [9], Threshold-Free OSR [8], Capsule+KLD [7].

4.4 Evaluation Metrics

Few-shot classification accuracy:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of query samples}} \times 100\%$$

Open-set evaluation: known class accuracy and unknown class accuracy.

4.5 Computational Environment

Hardware: Intel Xeon Gold 6230, NVIDIA Tesla V100 (32GB), 128GB RAM. Software: Ubuntu 20.04, Python 3.8.10, PyTorch 1.10.0, CUDA 11.3.

Chapter 5

Results and Analysis

5.1 Ablation Studies

Table 5.1 has the results from our ablation study on MSTAR and SAMPLE, for both 3-way 1-shot and 5-shot tests. What we found was that if you take away any of these components, things get measurably worse, showing us that every single module is necessary.

Table 5.1: Ablation Study Performance Under Few-Shot Settings

Absent Module	**3-way 1-shot (%)**		**3-way 5-shot (%)**	
	MSTAR	SAMPLE	MSTAR	SAMPLE
Patchification	81.53	80.42	86.28	84.50
Entropy Filtering	84.39	82.56	88.25	86.54
Cluster Consistency	86.61	84.59	90.16	88.67
Dependency Matrix	87.25	85.32	91.08	89.92
Proposed (Full)	**91.12**	**90.58**	**95.92**	**94.26**

5.2 Comparison with State-of-the-Art Approaches

Table 5.2 lets us see how the framework we're suggesting stacks up against five of the best ways to do this. This approach is more accurate across all the different shot setups in both datasets.

5.3 Open-Set Recognition Evaluation

Looking at Table 5.3, you'll see the results for open sets. Our suggested approach does a better job recognizing unknown classes, and it still keeps its accuracy high for the ones it already knows.

Table 5.2: Benchmark Comparison on Imbalanced MSTAR and SAMPLE Few-Shot Classification

Method	**MSTAR**				**SAMPLE**			
	K=1	K=3	K=5	K=7	K=1	K=3	K=5	K=7
ConvT [16]	77.76	79.78	81.05	82.43	76.12	80.82	82.15	82.56
Meta-FSCIL [11]	81.09	83.34	83.82	84.54	78.05	81.15	82.83	86.79
Position-Aware GNN [13]	81.76	86.15	87.56	89.87	79.70	83.07	85.78	89.50
MHD-ProtoNet [15]	88.67	90.56	91.87	92.06	88.09	88.53	90.59	90.72
MCL-DMM [14]	90.79	92.17	93.05	93.97	89.73	91.55	91.82	92.75
Proposed	**91.12**	**94.84**	**95.92**	**96.24**	**90.58**	**92.95**	**94.26**	**94.89**

Table 5.3: Open-Set Recognition Performance Comparison

Method	**Known Class Acc. (%)**		**Unknown Class Acc. (%)**	
	MSTAR	SAMPLE	MSTAR	SAMPLE
Incremental OSR [9]	90.32	88.76	73.71	69.93
Threshold-Free OSR [8]	89.10	87.45	70.60	68.97
Capsule + KLD [7]	88.87	89.12	71.89	69.28
Proposed	**91.42**	**90.01**	**79.81**	**74.56**

5.4 Qualitative Analysis

Figure 5.1, As shown by the figure above, samples within the same class have formed tight and distinguishable clusters, thus suggesting high class-consistency between similar samples. Additionally, it can be observed that classes are highly separable from each other and do not exhibit any overlap in the feature space, thus confirming high class-discriminability. As can be seen from the figure above, the results remain consistent across multiple cases, which includes balanced and imbalanced distributions, as well as unseen data points.

Notably, unknown or novel classes do not get embedded into the pre-existing clusters. Rather, they form new clusters on their own, which further confirms that the feature distributions have been captured in a meaningful manner. This ensures that the model does not make incorrect predictions due to any sort of overconfidence in its decisions.

On the whole, the use of the t-SNE algorithm confirms that the framework presented above successfully creates a structured feature space that is composed of tightly clustered known classes and distinct unknown samples.

Chapter 6

Conclusion and Future Work

6.1 Summary of Contributions

In this paper, we present an approach to detect SAR targets under conditions of scarce training samples, discriminate unknown classes of targets and at the same time guarantee the reliability of our system. In our research, there are four contributions proposed: one is that we propose a way of choosing discriminative regions from SAR images using a feature called 'entropy', another is that we construct a target template from the 'dependency matrix', moreover, we have developed a reliable episodic learning algorithm, and last but not least, a flexible technique of rejecting unknown targets. Our experiments conducted on the MSTAR database and SAMPLE databases have demonstrated our approach is comparable with the state-of-the-art approaches.

6.2 Interpretation of Results

During the analysis of the model components, it became clear that all the elements play a significant role in determining its efficiency. Thus, the approach to selecting image patches with respect to their entropy plays an especially important role in the case of working with scarce categories of images since without the element, the model's accuracy will decrease by approximately 6-8%. Furthermore, the dependency matrix provides the next step toward enhancing the overall performance of the model by providing us with 4-5% gain in efficiency since it makes it possible to determine dependencies between channels while centroid prototypes cannot do that. Lastly, the design of the loss function with various terms helps ensure the quality of the feature space organization.

6.3 Comparison with Existing Methods

Our method outperforms existing techniques by an average of 2-15%. The superiority of the proposed solution highlights the importance of considering the challenges of small sample sizes, imbalanced training datasets, and novel objects in combination with each other. Previous algorithms, such as ConvT, lack intrinsic mechanisms for self-assessment regarding their prediction quality. In contrast, prototype-based solutions like MCL-DMM perform better since they impose additional constraints on the processing procedure. For object recognition in completely novel contexts, our rejection-based technique utilizing dynamic thresholding of novelty score leads to performance improvements by 6-11%. This performance gain is achieved in comparison with alternative strategies, which rely either on artificial generation of unknown cases or require extensive human intervention.

6.4 Limitations

- A constant threshold value such as k=3 may not always be the best for all cases where the distribution of data classes is varied.
- As the dependency matrix increases, the number of dimensions in the prototypes also increases, hence increasing the processing power required by the computer.
- Using just one patch size may not always be the best decision because of the varied nature of target sizes.
- Our evaluation was quite limited; we only used two datasets, and both were just about identifying targets.
- We didn't fully explore all the possibilities that synthetic SAR data could offer.

6.5 Directions for Future Work

1. **Adaptive Threshold Optimization:** In open-set classification, the threshold is usually determined on a per-class basis. There is a need for future research to investigate adaptive or learned methods of determining the threshold.
2. **Efficient Dependency Matrix Representations:** Explore techniques for achieving low-rank approximations, using random projections, or learning of subspaces that can help in dimensionality reduction of dependency matrices.

3. **Robustness to Speckle and Noise:** Design training approaches and priors that will help models become more robust to SAR-related artifacts like speckle noise, thermal noise, and azimuth ambiguities.

4. **Cross-Domain and Cross-Sensor Generalization:** Consider employing domain adaptation methodologies in order to apply the representations gained from one SAR sensor to another one.

5. **Uncertainty Quantification:** Integrate Bayesian or conformal prediction frameworks to provide calibrated confidence estimates alongside classification decisions, which is critical for safety-sensitive applications.

6. **Multi-Scale and Adaptive Patch Selection:** Develop multi-scale patch analysis and learnable region proposals to better capture target structures at varying resolutions and aspect angles.

7. **Integration with Self-Supervised Pretraining:** Leverage large quantities of unlabeled SAR data through contrastive pretraining or masked autoencoding to improve feature representations with minimal supervision.

8. **Extension to Multi-Modal Learning:** Incorporate SAR-optical fusion, polarimetric SAR, or InSAR data to enrich scene understanding and improve classification accuracy across diverse target types.

9. **Incremental and Continual Learning:** Develop mechanisms to incrementally add new target classes without catastrophic forgetting of previously learned categories.

10. **Explainability Enhancement:** Provide patch-level explanations and visualize channel importance to improve model interpretability and support analyst trust in operational settings.

Publications

Title: A novel reliability aware patch selection based few shot learning for target recognition and open-set identification in ill posed SAR images
Authors: Anisha Chakraborty, Abhishek , Shounak Chakraborty , Abhishek Soni
Conference: IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026
Date: December, 2025
Status: Accepted for publication in IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026.
Link: Click here to View the Paper

Significance:

- Recognized by a top-tier international conference in remote sensing.
- Contributes to advancing solutions for class imbalance in aerial imagery segmentation.

Bibliography

[1] A. Tsokas, M. Rysz, P. M. Pardalos, and K. Dipple, "Sar data applications in earth observation: An overview," *Expert Systems with Applications*, vol. 205, p. 117342, 2022.

[2] X. X. Zhu, S. Montazeri, M. Ali, Y. Hua, Y. Wang, L. Mou, Y. Shi, F. Xu, and R. Bamler, "Deep learning meets sar: Concepts, models, pitfalls, and perspectives," *IEEE Geoscience and Remote Sensing Magazine*, vol. 9, no. 4, pp. 143–172, 2021.

[3] X. Huang, B. Zhang, W. Perrie, Y. Lu, and C. Wang, "A novel deep learning method for marine oil spill detection from satellite synthetic aperture radar imagery," *Marine Pollution Bulletin*, vol. 179, p. 113666, 2022.

[4] P. Chen et al., "Class imbalance in the automatic interpretation of remote sensing images," *IEEE Transactions on Geoscience and Remote Sensing*, 2025.

[5] N. Firoozy et al., "Tackling sar imagery ship classification imbalance via deep convolutional generative adversarial network," *International Journal of Remote Sensing*, 2021.

[6] Y. Zhang, Z. Lei, H. Yu, and L. Zhuang, "Imbalanced high-resolution sar ship recognition method based on a lightweight cnn," *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2021.

[7] C. Jiang, H. Zhang, R. Zhan, W. Shu, and J. Zhang, "Open-set recognition model for sar target based on capsule network with the kld," *Remote Sensing*, vol. 16, no. 17, p. 3141, 2024.

[8] Y. Li, H. Ren, X. Yu, C. Zhang, L. Zou, and Y. Zhou, "Threshold-free open-set learning network for sar automatic target recognition," *IEEE Sensors Journal*, vol. 24, no. 5, pp. 6700–6710, 2024.

[9] X. Ma, K. Ji, S. Feng, L. Zhang, B. Xiong, and G. Kuang, "Open set recognition with incremental learning for sar target classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–14, 2023.

[10] Y. Ma, X. Zhang, J. Pei, W. Huo, Y. Zhang, Y. Huang, and J. Yang, "A novel sar target recognition approach under imbalanced categories: Constraint and optimization," in *IGARSS 2024*, pp. 9644–9654, 2024.

[11] H. Ran, W. Li, L. Li, S. Tian, X. Ning, and P. Tiwari, "Learning optimal inter-class margin adaptively for few-shot class-incremental learning via neural collapse-based meta-learning," *Information Processing & Management*, vol. 61, no. 3, p. 103664, 2024.

[12] L.-J. Zhao, Z.-D. Chen, Z.-C. Zhang, X. Luo, and X.-S. Xu, "Bias mitigating few-shot class-incremental learning," *arXiv preprint*, 2024.

[13] J. Zheng, M. Li, P. Zhang, Y. Wu, and H. Chen, "Position-aware graph neural network for few-shot sar target classification," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 17, pp. 8028–8042, 2024.

[14] H. Tan, Z. Zhang, X. Shi, X. Yang, Y. Li, X. Bai, and F. Zhou, "Few-shot sar atr via multi-level contrastive learning and dependence matrix-based measurement," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 18, pp. 8175–8188, 2025.

[15] M. Zayani, A. Toumi, and A. Khalfallah, "Mhd-protonet: Margin-aware hard example mining for sar few-shot learning via dual-loss optimization," *Algorithms*, vol. 18, no. 8, p. 519, 2025.

[16] C. Wang, Y. Huang, X. Liu, J. Pei, Y. Zhang, and J. Yang, "Global in local: A convolutional transformer for sar atr fsl," *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2022.

[17] T. D. Ross, S. W. Worrell, V. J. Velten, J. C. Mossing, and M. L. Bryant, "Standard sar at evaluation experiments using the mstar public release data set," in *Algorithms for synthetic aperture radar imagery V*, vol. 3370, pp. 566–573, SPIE, 1998.

[18] B. Lewis, T. Scarnati, E. Sudkamp, J. Nehrbass, S. Rosencrantz, and E. Zelnio, "A sar dataset for art development: the synthetic and measured paired labeled experiment (sample)," in *Algorithms for Synthetic Aperture Radar Imagery XXVI*, vol. 10987, pp. 39–54, SPIE, 2019.